

SMART LIFE: AN ESP32-BASED IoT PLATFORM FOR INTEGRATED HOME AUTOMATION

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BONAFIDE CERTIFICATE

Certified that this mini project report entitled “SMART LIFE: AN ESP32-BASED IoT PLATFORM FOR INTEGRATED HOME AUTOMATION” is the bonafide work submitted by the following students of the Department of Artificial Intelligence and Machine Learning, School of Engineering, Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai, under the guidance of S. Hariprabha.

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The report presents the design and prototype of Smart Life, an ESP32-based home-automation system that combines a local web dashboard, relay-based appliance control, environmental sensor monitoring, and infrared remote control.

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ABSTRACT

Smart home systems can make appliance control difficult when users have to manage different devices separately. They may also lack one convenient place to monitor the home and respond to certain safety-related situations. This project presents Smart Life, an IoT-based home automation system developed using an ESP32 DevKit. It brings appliance control, environmental monitoring, and basic safety functions together in one system. Users access a web dashboard over a local Wi-Fi network to control appliances connected through a four-channel relay module and operate compatible infrared (IR) devices, including televisions and air conditioners. A PIR sensor monitors motion, an LDR senses changes in ambient light, and an MQ-2 sensor provides gas-related readings. When the configured gas threshold is exceeded, the system is designed to activate the assigned relay, flash an LED, and show an alert on the dashboard with an audible indication where configured. Other described features include IR remote learning and replay, user login, CAPTCHA-based login protection, OTP-based account recovery, and typed or voice commands. The dashboard runs locally, so paid cloud hosting is not required for local operation. The supplied project materials describe a prototype integrating embedded hardware, sensors, and wireless communication. Further testing is needed to measure sensor accuracy, response time, reliability, and security in different operating conditions.

Keywords: Internet of Things (IoT); ESP32; smart home automation; web dashboard; PIR sensor; LDR sensor; MQ-2 gas sensor; infrared remote control; wireless communication; embedded systems.

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ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
CAPTCHA	Completely Automated Public Turing test to tell Computers and Humans Apart
ESP32	ESP32 microcontroller family
GPIO	General-Purpose Input/Output
HTTP	Hypertext Transfer Protocol
IR	Infrared
IoT	Internet of Things
LDR	Light-Dependent Resistor
MQ-2	Gas/smoke sensor module designation used in the project
OTP	One-Time Password
PIR	Passive Infrared
PWM	Pulse-Width Modulation
REST	Representational State Transfer
VISTAS	Vels Institute of Science, Technology and Advanced Studies
Wi-Fi	Wireless Fidelity

CHAPTER 1. INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Home automation is intended to make everyday appliance control easier and to provide a convenient way to monitor selected conditions in a home. When lights, fans, televisions, and air conditioners rely on separate switches or remotes, users may need to move between different controls. Basic environmental alerts may also be separate from appliance control.

Smart Life is an ESP32-based prototype that combines a local Wi-Fi web dashboard, relay-based appliance switching, infrared control, and environmental sensing. The ESP32 DevKit hosts a web server that communicates with a browser on the same home Wi-Fi network. A four-channel relay module is used to switch connected loads, while an IR receiver and IR transmitter support learning and replaying compatible remote-control signals.

The project includes a PIR motion sensor, an LDR light sensor, and an MQ-2 gas sensor. The MQ-2 reading is compared with a configured threshold; when the threshold is exceeded, the safety logic is described as activating a relay response, blinking an indicator LED, and displaying an alert on the dashboard. The design aims to keep these functions available from one interface without requiring external cloud hosting for local control.

1.2 PROBLEM STATEMENT

Managing household appliances through separate switches and remotes can be inconvenient. A user may also lack a single place to check sensor conditions and receive a clear indication of a gas-related threshold event. Some commercial systems depend on external cloud accounts or vendor-specific applications.

The project addresses this problem by developing a local dashboard that combines relay switching, IR-based control, and basic sensor monitoring using an ESP32. It also aims to provide a threshold-based safety indication and a unified view of connected devices.

1.3 OBJECTIVES

- Develop a home-automation prototype using an ESP32 DevKit and Wi-Fi.
- Provide a local browser dashboard for appliance control and status viewing.
- Switch connected appliances using the four-channel relay module.
- Integrate PIR, LDR, and MQ-2 sensor inputs for motion, light, and gas-related monitoring.
- Support learning and retransmission of IR commands for compatible televisions and air conditioners.
- Provide a dashboard warning and indicator response when the configured gas threshold is exceeded.
- Include the access and command features described in the project materials, subject to validation in the final build.

1.4 SCOPE OF THE PROJECT

The supplied project materials describe a prototype intended for a home or small research setup. Its main scope includes local Wi-Fi operation, dashboard-based relay control, sensor inputs, IR learning/replay, and gas-threshold alerts. The ESP32 serves the dashboard locally; external cloud hosting is not required for the basic described operation.

The report includes the hardware and software design, sensor and relay roles, the documented endpoint patterns, and the evidence shown in the project presentation. Features that lack numerical evaluation are described without invented accuracy percentages, response times, or reliability claims.

The prototype is not presented as a certified gas-safety product. Any connection to mains-powered appliances must use appropriate isolation, protective enclosures, rated components, and qualified supervision.

1.5 APPLICATIONS AND EXPECTED OUTCOMES

- Centralized control of relay-connected lights, fans, and other compatible loads.
- Browser-based operation from a device connected to the same Wi-Fi network.

- IR control of compatible televisions and air conditioners.
- Monitoring of motion, light level, and MQ-2 gas-related readings.
- Local visual and dashboard indication when the configured gas threshold is exceeded.
- An educational platform for demonstrating embedded systems, sensors, Wi-Fi communication, and home automation.

The expected outcome is a working prototype that demonstrates integration of these modules. Larger installations and long-term reliability require additional testing.

CHAPTER 2. LITERATURE REVIEW

2.1 BACKGROUND

Smart-home systems commonly combine a controller, communication network, device outputs, and a user interface. Existing approaches described in the supplied conference paper often focus on relay switching over Wi-Fi or Bluetooth through a mobile application or web page. Environmental alerts may be treated as a separate feature rather than being integrated with everyday appliance control.

Gas-monitoring systems using MQ-series sensors commonly compare an analog reading with a configured threshold and trigger a buzzer, relay, or other warning output. Infrared remote-control systems use a receiver to capture commands and a transmitter to replay them to compatible appliances. Voice-command interfaces can provide another way to submit commands, but the recognized command must be mapped to an existing control function.

2.2 EXISTING APPROACHES

The supplied conference paper describes existing ESP-based home-automation systems that commonly focus on relay switching through Wi-Fi or Bluetooth using a mobile application or web page. It also notes that gas-sensor alerts are often treated as a secondary function, while MQ-series sensors are used for LPG or smoke detection by comparing an analog reading with a threshold and activating a buzzer or relay. Infrared remote replacement is another approach discussed in the paper, but appliance compatibility depends on the supported protocol and the actual remote. The paper positions Smart Life as an integration of these functions in a locally hosted dashboard.

APPROACH	MAIN IDEA	LIMITATION / CONSIDERATION
Wi-Fi or Bluetooth appliance control	Use an app or web interface to switch appliances.	Control and safety monitoring may remain separate.
Sensor-based gas alert	Read a gas sensor and trigger an alarm at a threshold.	Needs calibration and testing under different conditions.
IR remote control	Transmit known infrared commands to consumer devices.	Compatibility depends on the appliance and IR protocol.
Voice or typed commands	Convert user commands into device-control actions.	Command parsing and access security need validation.
Cloud-hosted smart-home systems	Use external servers or vendor platforms.	May require internet access or an external account for some functions.

2.3 RESEARCH GAP

The supplied paper identifies a practical need to combine appliance control, sensor-based safety indication, and IR control within one interface. A system that only switches relays may not provide environmental monitoring. A separate sensor alarm may not be visible in the appliance dashboard. A fixed-code IR controller may also be difficult to adapt to different remotes.

Smart Life addresses this integration goal by combining a locally hosted dashboard with relay control, PIR/LDR/MQ-2 inputs, and IR learning/replay. The supplied sources do not provide a controlled benchmark against commercial systems or quantitative performance measurements.

2.4 PROPOSED CONTRIBUTION

- One local dashboard for relay-connected appliances and compatible IR-controlled devices.
- ESP32-hosted web server and Wi-Fi communication without cloud hosting for local operation.
- Integration of PIR motion sensing, LDR light sensing, and MQ-2 gas-related monitoring.
- Threshold-based gas-alert logic linked to a relay response, LED indication, and dashboard warning.
- IR remote learning/replay and the account/command features described in the project materials.

The contribution is the prototype-level integration of these modules. The report does not claim unmeasured gains in speed, accuracy, cost, or safety.

CHAPTER 3. SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

Smart Life has a local user-interface layer and an embedded-control layer. The browser dashboard sends commands to the ESP32 over the home Wi-Fi network. The ESP32 web server receives requests, applies the corresponding control logic, and updates relay, sensor, or IR functions. Sensor values and device status can then be shown in the dashboard.

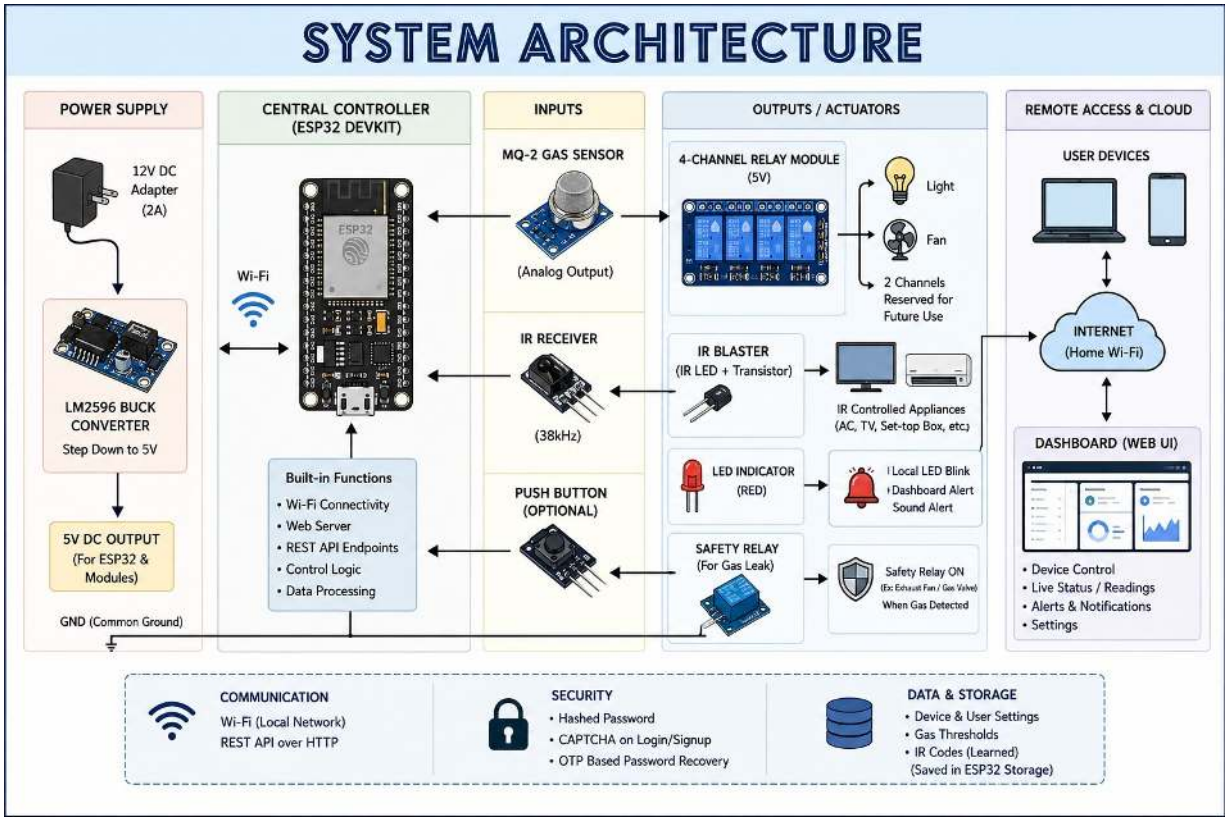


Figure 1 – System architecture reproduced from the supplied Smart Life presentation.

LAYER	MAIN COMPONENT	ROLE
User interface	Browser-based Smart Life dashboard	Displays device cards, controls, and status.
Network	Home Wi-Fi	Connects the browser device and ESP32.
Controller	ESP32 DevKit and local web server	Processes requests and reads sensor inputs.
Actuation	Four-channel relay module and IR transmitter	Switches connected loads or retransmits IR commands.
Monitoring	PIR, LDR, MQ-2, LED and buzzer where fitted	Provides input readings and safety indications.

3.1.1 OVERALL DATA FLOW

1. The user opens the dashboard from a browser on the same home Wi-Fi network.
2. The dashboard submits a relay, IR, or configuration request to the ESP32 web server.
3. The ESP32 validates the request and calls the matching control routine.
4. The selected relay output changes, or an IR command is retransmitted to a compatible appliance.
5. The ESP32 reads sensor inputs and compares the MQ-2 reading with the configured threshold.
6. When the gas threshold is exceeded, the described logic activates the safety response and displays an alert.
7. The dashboard presents device status and relevant sensor/alert information.

Figure 2 is a simplified logical flow derived from the architecture described in the paper; it is not a screenshot of the firmware.

STEP	ACTION
1	Browser dashboard → local Wi-Fi request
2	ESP32 web server receives and validates request
3	Control logic → relay / IR transmitter / sensor handler
4	Sensor reading and threshold evaluation
5	Dashboard status or safety alert updated

Figure 2 – Data flow from the dashboard or sensors to the control outputs

3.2 HARDWARE COMPONENTS

COMPONENT	ROLE IN THE PROTOTYPE
ESP32 DevKit	Central controller; Wi-Fi connection and local web server.
Four-channel relay module	Switching connected appliance loads.
PIR motion sensor	Detects motion; the project configuration associates the PIR input with relay-control behaviour.
LDR light sensor	Reads ambient light level for the configured light-related logic.
MQ-2 gas sensor	Provides an analog gas/smoke-related reading for threshold monitoring.
IR receiver	Captures signals from a physical remote.
IR LED transmitter	Retransmits learned IR commands to compatible devices.
Buzzer	Audible indication where included in the implemented build.
LED and resistor	Visual indication for a gas-alert event.
Regulated power supply / converter	Supplies the controller and peripheral modules within their ratings.
Jumper wires and prototype base	Provide electrical connections and physical support.

3.3 PIN CONFIGURATION

The following assignments are taken from the project details previously supplied for Smart Life. They must be checked against the final sketch and physical wiring before powering the circuit.

COMPONENT	GPIO / CONNECTION	FUNCTION
Relay 1	GPIO 16	Relay output for connected load.
Relay 3	GPIO 18	Relay output; also referenced in the PIR-related behaviour.
Relay 4	GPIO 23	Relay output associated with the LDR-related behaviour.
PIR sensor	GPIO 27	Motion input.
LDR sensor	GPIO 35	Analog light-level input.
MQ-2 gas sensor	GPIO 34	Analog gas-related input.
Buzzer	GPIO 25	Audible alert output.
LED	GPIO 2	Visual alert output.
IR receiver	GPIO 21	IR signal input.
IR LED transmitter	GPIO 22	IR signal output.

A pin for Relay 2 is not listed in the supplied pin details, so it is intentionally not inferred here. GPIO availability and analog-input behaviour should be verified for the exact ESP32 board variant.

3.4 SOFTWARE ARCHITECTURE

The conference paper describes ESP32 firmware written in Arduino-style C/C++ with Wi-Fi and HTTP server handling, relay functions, IR learning/replay, sensor logic, and persistent configuration. The browser dashboard uses HTML, CSS, and JavaScript and communicates with the firmware through REST-style HTTP endpoints.

SOFTWARE ELEMENT	PURPOSE

Arduino-compatible ESP32 firmware	Initializes Wi-Fi, sensors, GPIO outputs and the local web server.
Wi-Fi / HTTP web-server libraries	Allow the dashboard to send commands to the ESP32.
HTML and CSS	Define the dashboard structure and visual layout.
JavaScript	Handles dashboard interactions and HTTP requests.
IR control library	Supports decoding and retransmitting compatible infrared signals; the conference paper names IRremoteESP8266.
Flash / EEPROM-style persistence	Stores configuration such as the gas threshold and saved device settings, as described in the source.
Web browser	Displays the dashboard on a device connected to the same network.

3.5 DASHBOARD AND API ENDPOINTS

The conference paper lists the following REST-style route patterns. The exact query parameters should be checked against the final firmware before they are used as a definitive API specification.

ROUTE PATTERN FROM PAPER	PURPOSE
/relay?pin=N&state=1|0	Switch a relay channel on or off.
/pwm?pin=N&value=0–255	Set a PWM value where the selected output and hardware support it.
/ir?code=NAME&state=1|0	Replay a learned IR command.
/ir/learn?name=NAME	Begin learning an IR button.
/ir/ac?power=…&temp=…&mode=…	Set an air-conditioner state through IR control.
/gas	Read the current gas-sensor value.
/gas/threshold	Read or change the gas alert threshold.
/status	Return current device states.

The dashboard groups device controls into cards and includes a command bar for typed or voice commands, as described in the supplied paper and presentation. Local operation depends on the browser device and ESP32 being reachable on the same Wi-Fi network.

CHAPTER 4. IMPLEMENTATION

4.1 HARDWARE IMPLEMENTATION

The prototype places the ESP32, relay module, sensor modules, IR components, and indicators on a stable base. The controller connects to the home Wi-Fi network and hosts the web server. Sensor inputs are read by the firmware; relay outputs and the IR transmitter respond to dashboard commands or the configured sensor logic.

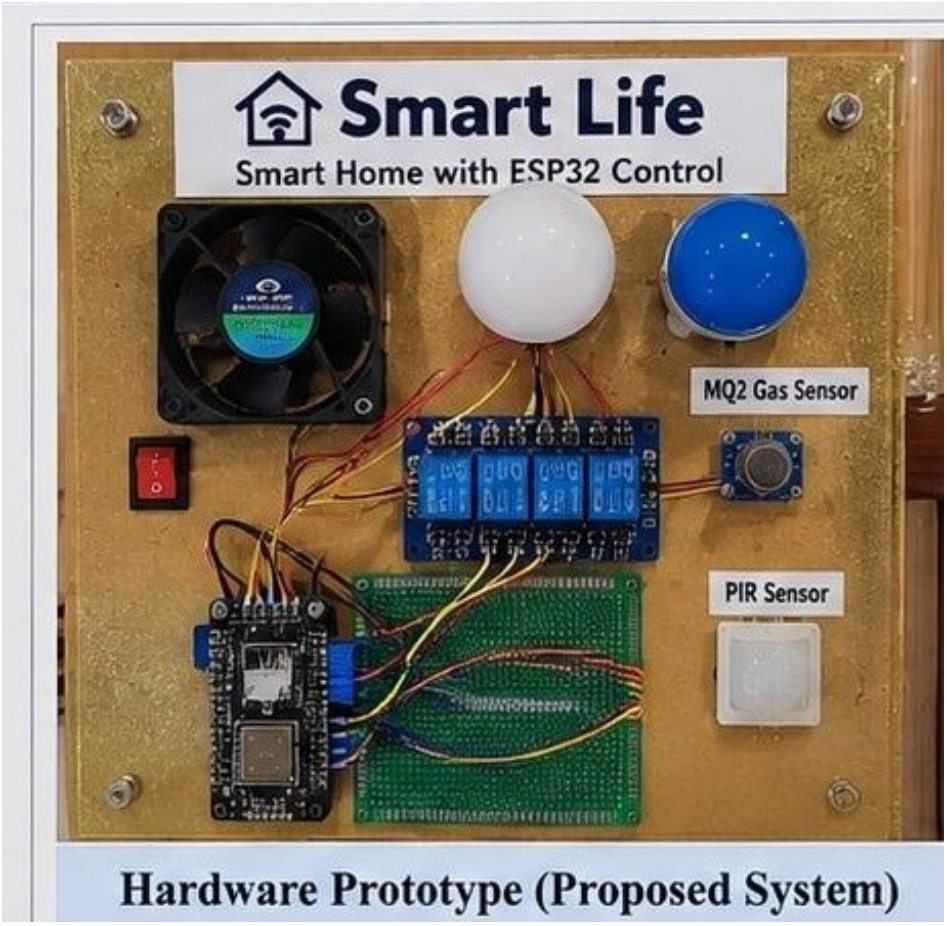


Figure 3 – Hardware prototype shown in the supplied Smart Life presentation.

The photograph is taken from the project presentation and is included as project evidence. The final report should use the clearest original photograph available from the team if a higher-resolution image exists.

4.1.1 POWER AND ELECTRICAL SAFETY

The ESP32 GPIO pins are not intended to drive high-current appliances directly. Relay modules and sensors must be powered within their rated voltage ranges, and the wiring must follow the requirements of the specific modules. Mains wiring should be enclosed and protected, and must be handled by a qualified person. Demonstrations should use low-voltage loads wherever possible.

4.2 DASHBOARD AND APPLIANCE CONTROL

The local dashboard presents appliance controls in a single browser interface. The project presentation shows cards for relay-controlled light and fan outputs and IR-based controls for compatible TV/AC devices. Relay buttons issue a state change to the ESP32; the firmware then updates the relevant GPIO output.



Figure 4 – Web dashboard with device tabs and voice-command interface, from the supplied presentation.

The command bar is described as accepting typed or voice commands such as turning a named appliance on or off or asking for status. The recognized command is mapped to the same control functions used by the dashboard buttons. Voice-command support should be demonstrated in the final build before being claimed as fully implemented.



Figure 5 – Appliance-control example for light and fan, from the supplied presentation.

4.3 SENSOR MONITORING AND SAFETY LOGIC

4.3.1 PIR MOTION SENSOR

The PIR sensor supplies a digital motion input. The project details associate motion detection with a relay-control action and a short hold interval. The exact timing and output should be confirmed against the final firmware and observed during the demonstration.

4.3.2 LDR LIGHT SENSOR

The LDR provides an input related to ambient light. The supplied project details associate the LDR with Relay 4. The threshold, polarity and behaviour depend on the sensor circuit and final firmware, so the report does not assign an unverified numeric light threshold.

4.3.3 MQ-2 GAS SENSOR

The MQ-2 module provides an analog gas/smoke-related reading. The firmware compares the reading with a configurable threshold. The described safety response activates the assigned relay, blinks an LED, and displays a warning on the dashboard with an audible indication where configured.



Figure 6 – MQ-2 gas sensor module shown in the supplied presentation.

An MQ-2 module requires appropriate warm-up, calibration, and controlled testing. This educational prototype must not replace a certified gas detector or a professionally installed safety system.

4.4 IR REMOTE LEARNING

The IR subsystem is designed to learn commands from a physical remote and replay them from the dashboard. The conference paper describes the following process:

8. Add a compatible TV or air conditioner to the dashboard.
9. Enable Learn mode for the desired device or button.
10. Point the physical remote at the ESP32 IR receiver and press the button once.
11. The firmware decodes the received signal and associates it with the selected control.
12. Turn Learn mode off and test the dashboard button to retransmit the stored command.

The paper also describes automatic protocol detection for air-conditioner remotes. Actual compatibility depends on the appliance, remote protocol, receiver placement, and firmware support. The term “universal” should not be interpreted as a guarantee that every IR-controlled appliance is supported.

4.5 USER ACCESS AND CONFIGURATION

The conference paper describes a username and hashed-password login, CAPTCHA on sign-in and sign-up, a six-digit OTP step for password recovery, and a “Keep me signed in” option. It also describes storing configuration such as the ESP32 address, device definitions, and gas-alert threshold.

These features must be demonstrated in the final software before they are all described as completed. Credentials, Wi-Fi passwords, OTP secrets, and other sensitive configuration values should not be included in a public report or source-code appendix.

4.6 METHODOLOGY

The implementation process is modular so that hardware, network, and dashboard issues can be isolated during testing.

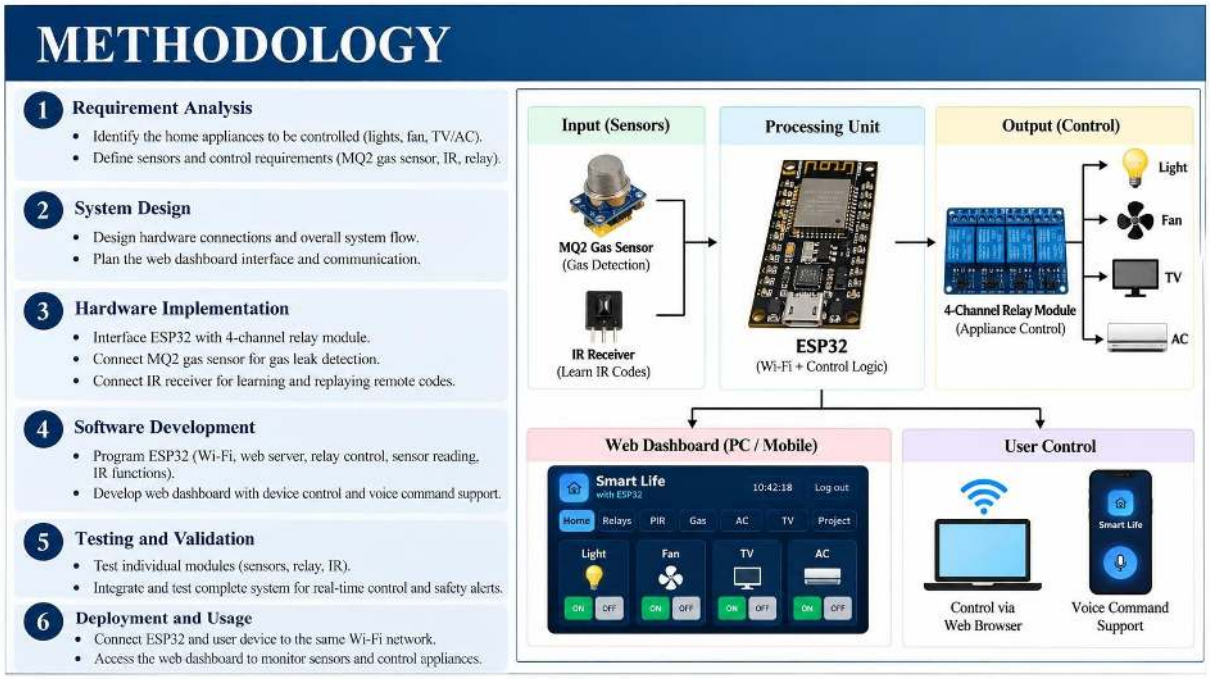


Figure 7 – Methodology infographic supplied in the project presentation.

13. Requirement identification: list appliances, sensor inputs, dashboard controls, and safety behaviour.
14. Hardware setup: connect the ESP32, relay module, sensors, indicators and IR components using the final pin map.
15. Firmware development: implement Wi-Fi, HTTP routes, GPIO control, sensor sampling, IR functions and configuration handling.
16. Dashboard development: provide browser controls, device status, and sensor/alert panels.
17. Integration: connect dashboard requests to firmware routines and link sensor thresholds to the documented response.
18. Testing: test each module separately and then run integrated demonstrations.
19. Documentation: capture actual screenshots and record observations without inventing performance metrics.

4.7 SOURCE CODE IMPLEMENTATION

The conference paper describes a web server running on port 80 and route handlers for relay switching, PWM control, IR learning/replay, air-conditioner state, gas readings, and overall status. The following is a short excerpt of the setup pattern shown in the supplied project presentation. It is not the complete firmware.

```
#include      <WiFi.h>\n#include      <WebServer.h>\n#include      <EEPROM.h>\n#include
<IRremote.h>\n\nconst char* ssid = "YourWiFi";\nconst char* password = "YourPassword";
\n\nWebServer server(80);
```

The SSID and password above are placeholders shown in the presentation; they are not the project team's actual credentials. The final firmware source was not supplied as a complete text file for this report, so no unverified full program is reproduced. The presentation screenshot shows an IRremote.h include, while the conference paper names IRremoteESP8266; confirm the library used by the final firmware before submission.

4.7.1 LOGIC SUMMARY

MODULE	LOGIC SUMMARY
Relay control	Receive the selected channel and requested state; validate the request; update the output; report the state.
Sensor monitoring	Read PIR, LDR and MQ-2 inputs and update the corresponding state variables.
Gas threshold	Compare the MQ-2 reading with the saved threshold; activate the configured relay and indicators when exceeded.
IR learning/replay	Capture an IR signal in Learn mode, associate it with a named control, then retransmit it on request.
Dashboard status	Provide device and sensor state to the browser through the documented status interface.

4.8 TESTING AND OBSERVATIONS

The supplied presentation includes a hardware prototype, a dashboard view, device-control examples, and sensor-related result panels. These visuals demonstrate the intended module arrangement and reported demonstration states. The presentation does not provide numerical measurements for sensor accuracy, update delay, or long-term reliability.

TEST CASE	EXPECTED / SHOWN BEHAVIOUR	EVIDENCE
Wi-Fi dashboard access	Browser dashboard is available through the ESP32's local network address.	Dashboard screenshot in presentation.
Relay appliance control	Light/fan output changes between OFF and ON states.	Device-control panel.
PIR sensor	Motion/no-motion states are shown in the presentation.	PIR sensor result panel.
LDR sensor	Bright/dark environment states are shown in the presentation.	LDR sensor result panel.
MQ-2 gas sensor	Normal/gas-detected states and indicator response are shown.	MQ-2 result panel.
IR control	A remote command is shown being sent to a TV.	IR control result panel.
IR learning/replay	The paper describes learning and retransmission of remote codes.	Confirm by live demonstration.
Login / OTP / CAPTCHA	Described in the paper and presentation.	Capture evidence from final working build.

4.9 RESULTS AND DISCUSSION

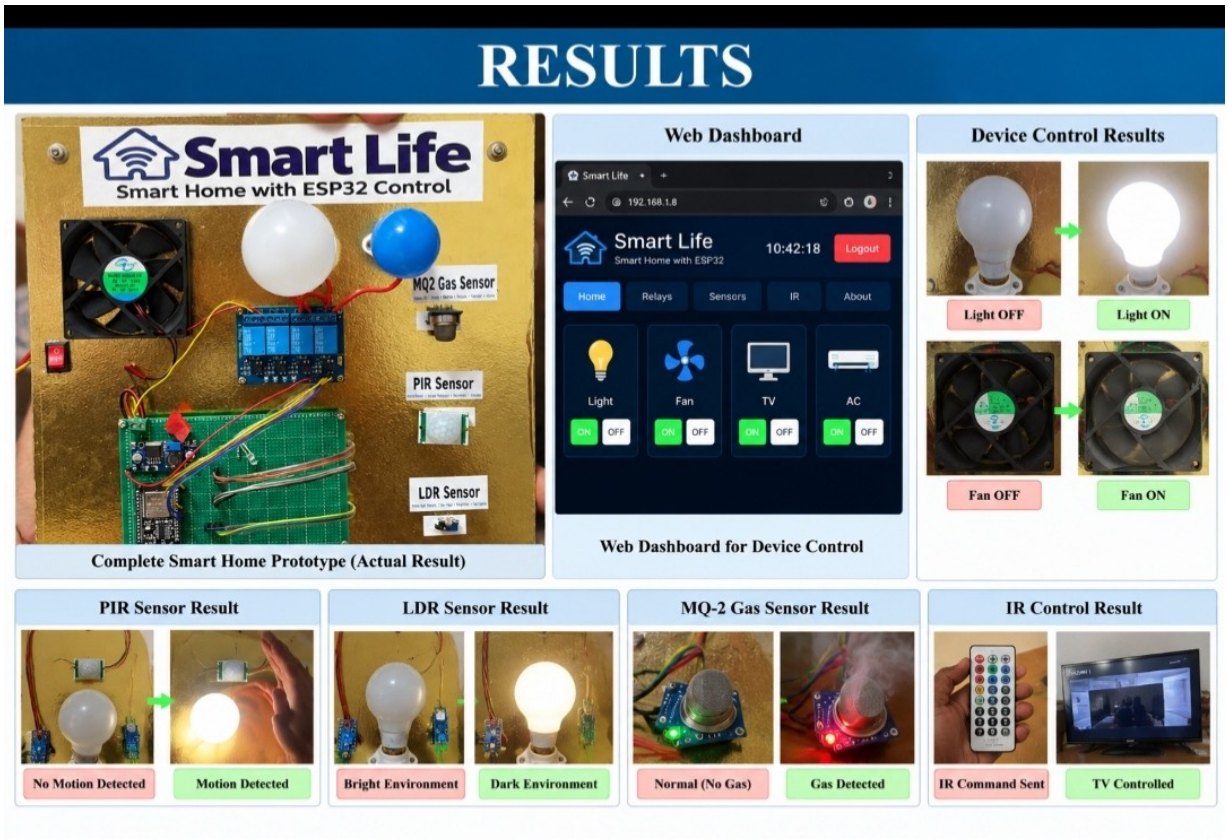


Figure 8 – Results collage supplied in the Smart Life presentation.

The supplied presentation reports the integration of a hardware prototype, a browser dashboard, relay-controlled light/fan outputs, PIR/LDR/MQ-2 sensor panels, and IR control. These visual results support a qualitative description of the prototype's functions.

The conference paper describes breadboard-level relay switching, IR learning/replay, and gas-alert behaviour. However, the presentation does not provide measured values for sensor accuracy, response time, or long-term reliability. No performance percentages or timing claims are added to this report.

A stronger evaluation would record the sensor reading before and after each test, the time from a physical event to the dashboard update, repeated relay cycles, false-alert cases, and the result of IR replay with each target appliance.

CHAPTER 5. CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

Smart Life is an ESP32-based home-automation prototype that brings relay switching, infrared control, and basic environmental monitoring into a local web dashboard. The project combines an ESP32 DevKit, a four-channel relay module, PIR/LDR/MQ-2 sensor inputs, IR learning/replay, and a threshold-based gas-alert routine. The local Wi-Fi design allows basic operation without external cloud hosting.

The project materials demonstrate the intended architecture and show prototype and dashboard evidence. They do not provide quantitative test results for sensor accuracy, update latency, security, or long-term reliability. Smart Life should therefore be described as an educational prototype that needs further controlled testing, not as a certified safety system.

5.2 LIMITATIONS

- The supplied presentation does not report numerical sensor-accuracy measurements.
- Response time and long-duration reliability have not been measured in the provided materials.
- IR compatibility depends on the target appliance and the signal supported by the firmware.
- Access-control and recovery features require validation in the final build.
- The gas-sensor response depends on calibration, placement, environmental conditions, and threshold configuration.
- The complete final firmware source file was not available as a text source for inclusion in this report.

5.3 FUTURE ENHANCEMENTS

- Perform repeated tests for relay operation, sensor calibration, gas-threshold behaviour, and false alerts.
- Record the time between a sensor event, ESP32 response, and dashboard update.
- Improve access security and validate password recovery, session handling, and request validation.
- Add further environmental sensors where required, after confirming the hardware and firmware design.
- Improve the dashboard's device feedback and error messages when the ESP32 or Wi-Fi is unavailable.
- Evaluate IR compatibility with the actual televisions and air conditioners used for demonstration.
- Document the complete firmware, dashboard files, library versions, and wiring diagram for reproducibility.

5.4 FINAL REMARK

The project shows how a low-cost microcontroller, a local browser interface, sensors, relay outputs, and infrared communication can be integrated into a home-automation prototype. Further testing and careful electrical safety practices are required before the system is considered for continuous or real-world use.

APPENDIX A – PIN MAP AND FIRMWARE NOTES

A.1 Pin map

DEVICE	GPIO	NOTE
Relay 1	16	Specified in project details.
Relay 3	18	Specified in project details.
Relay 4	23	Specified in project details; LDR-related behaviour is described.
PIR	27	Motion input.
LDR	35	Light-level input.
MQ-2	34	Analog gas-related input.
Buzzer	25	Audible indicator.
LED	2	Visual indicator.
IR receiver	21	IR input.
IR LED transmitter	22	IR output.

A.2 Endpoint notes

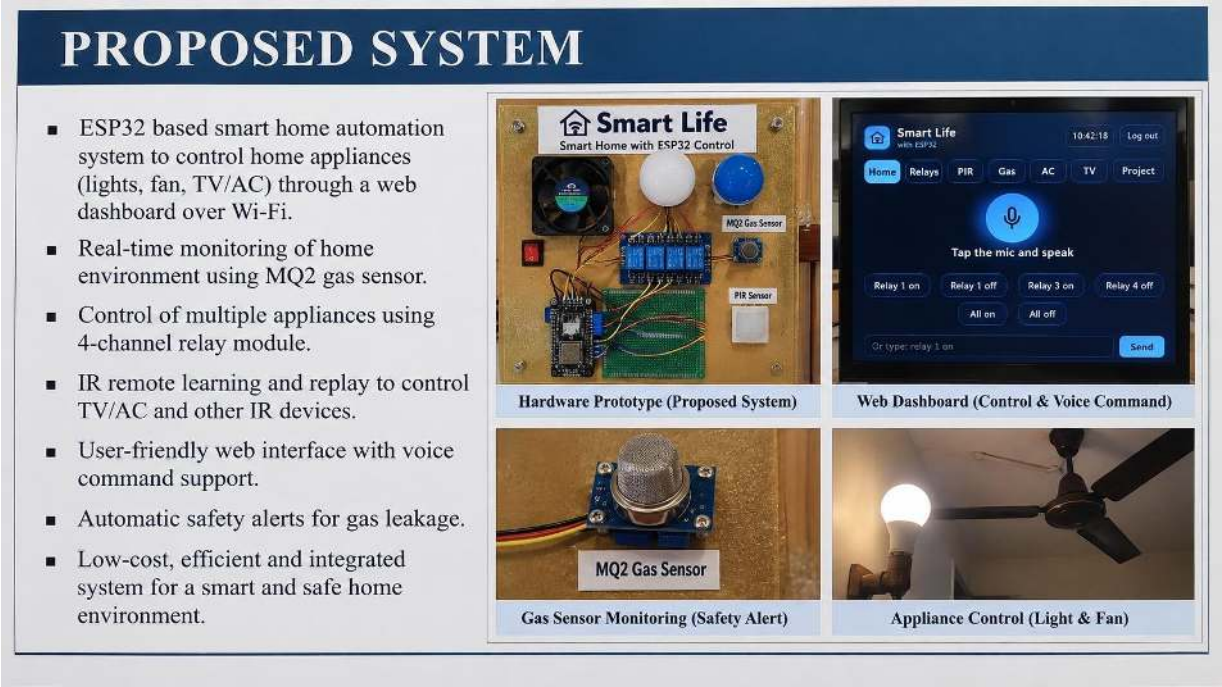
The endpoint names and parameter examples in Chapter 3 are taken from the supplied conference paper. Confirm exact paths and parameter names against the final code before using them in a live demonstration.

A.3 Security note

Replace all example Wi-Fi values with local settings only in the private development environment. Do not include real passwords, tokens, OTP secrets, or other credentials in the submitted report.

APPENDIX B – PROJECT PRESENTATION EVIDENCE

The images in this appendix are taken from the project team's supplied presentation. They are included as visual evidence of the project design and reported demonstration. Replace them with higher-resolution original photographs or screenshots if available.



Appendix Figure B1 – Proposed system overview and project visuals from the presentation.

The panel above shows the prototype board, dashboard interface, MQ-2 sensor module, and an appliance-control example. The report uses the presentation's own figures and does not claim these images are independent laboratory measurements.

APPENDIX C – CONFERENCE PARTICIPATION CERTIFICATES

Certificate 1 – Mukesh Kumar B.



SIMATS ENGINEERING

NBA Tier 1 | IET-UK | ABET Accreditation

PLATINUM

CERTIFICATE OF PARTICIPATION

This certificate is presented to

MUKESH KUMAR.B

VELS INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)

for actively participating in **AICTE Sponsored International Conference** on **Artificial Intelligence-Driven Innovations in Computer, Electrical, Mechanical and Electronics Engineering (ICAI-CEME 2026)** and presented the Research Paper titled

SMART LIFE: AN ESP32-BASED IOT PLATFORM FOR INTEGRATED HOME AUTOMATION

held during 21.10.2026 & 22.10.2026 organized by the Institute of CSE, SIMATS Engineering, SIMATS, Thandalam, Chennai. Your presence and engagement have contributed significantly to the discussions and knowledge-sharing on AI-driven Engineering Innovations.

Conference Chair
ICAI-CME 2026

Principal
Dr. B. Ramesh

Certificate 2 – Hari Prabha S.



SIMATS ENGINEERING

NBA Tier 1 | IET-UK | ABET Accreditation

PLATINUM

CERTIFICATE OF PARTICIPATION

This certificate is presented to

HARI PRABHA.S

VELS INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)

for actively participating in **AICTE Sponsored International Conference** on **Artificial Intelligence-Driven Innovations in Computer, Electrical, Mechanical and Electronics Engineering (ICAI-CEME 2026)** and presented the Research Paper titled

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Conference Chair
ICAI-CME 2026

Principal
Dr. B. Ramesh

Certificate 3 – Syed Mohammed Buhari S.



SIMATS ENGINEERING

NBA Tier 1 | IET-UK | ABET Accreditation

PLATINUM

A++ GRADE

45 RANK

GOALS

CERTIFICATE OF PARTICIPATION

This certificate is presented to

SYED MOHAMMED BUHARI.S

VELS INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)

for actively participating in **AICTE Sponsored International Conference** on **Artificial Intelligence-Driven Innovations in Computer, Electrical, Mechanical and Electronics Engineering (ICAI-CEME 2026)** and presented the Research Paper titled

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Conference Chair
ICAI-CME 2026

Principal
Dr. B. Ramesh

Certificate 4 – Yuvaraj Subramani M.



SIMATS ENGINEERING

NBA Tier 1 | IET-UK | ABET Accreditation

PLATINUM

A++ GRADE

45 RANK

GOALS

CERTIFICATE OF PARTICIPATION

This certificate is presented to

YUVARAJ SUBRAMANI.M

VELS INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)

for actively participating in **AICTE Sponsored International Conference** on **Artificial Intelligence-Driven Innovations in Computer, Electrical, Mechanical and Electronics Engineering (ICAI-CEME 2026)** and presented the Research Paper titled

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Conference Chair
ICAI-CME 2026

Principal
Dr. B. Ramesh

REFERENCES

[1] R. Piyare and M. Tazil, “Bluetooth based home automation system using cell phone,” in Proc. IEEE 15th International Symposium on Consumer Electronics (ISCE), 2011, pp. 192–195. (Bibliographic details reproduced from the supplied conference paper; verify against the original publication.)

[2] K. Sonar, S. Baviskar, and V. Panchal, “IoT based home automation system over cloud,” International Journal of Computer Applications, vol. 170, no. 5, pp. 1–5, 2017. (As listed in the supplied conference paper; verify against the original publication.)

[3] S. Saha, A. Majumdar, and S. Pal, “Gas leakage detection and alert system using IoT,” International Journal of Engineering Research & Technology, vol. 6, no. 5, pp. 210–213, 2018. (As listed in the supplied conference paper; verify against the original publication.)

[4] A. Kumar and R. Verma, “Universal IR remote controller using microcontroller,” International Journal of Computer Science and Engineering, vol. 5, no. 3, pp. 45–49, 2017. (As listed in the supplied conference paper; verify against the original publication.)

[5] P. Sharma and N. Gupta, “Voice controlled home automation system using IoT,” International Journal of Scientific Research in Computer Science and Engineering, vol. 6, no. 4, pp. 12–16, 2018. (As listed in the supplied conference paper; verify against the original publication.)

[6] “Smart Life: An ESP32-Based IoT Platform for Integrated Home Automation,” conference paper supplied by the project team for ICAI-CEME 2026, Paper ID ICAI26_ON_247.

[7] “Smart Life: An ESP32-Based IoT Platform for Integrated Home Automation,” project presentation supplied by the project team, ICAI-CEME 2026, 21–22 October 2026.

[8] “Smart Life: IoT-Based Smart Home Automation Using ESP32,” separate abstract supplied by the project team.